

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 6 — Work and Energy

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

1. C — Joule 1 mark

Work is defined as $W = Fd$ (force times displacement in the direction of force). Its SI unit is the **joule (J)**, where $1 \text{ J} = 1 \text{ N}\cdot\text{m}$. Watt is the unit of power, pascal is pressure, and newton is force.

2. C — Zero 1 mark

Work done $W = F \times d$. Since the wall does not move, displacement $d = 0$, therefore $W = 0$. No matter how large the force, work is only done when there is displacement in the direction of the force. The man uses energy (metabolic), but does no mechanical work on the wall.

3. B — 25 J 1 mark

$E_p = mgh$. Weight $W = mg = 5 \text{ N}$, height $h = 5 \text{ m}$. Therefore $E_p = 5 \text{ N} \times 5 \text{ m} = \mathbf{25 \text{ J}}$. (The weight in newtons already equals mg , so we use $E_p = W \times h = 5 \times 5 = 25 \text{ J}$.) Common error: multiplying mass in kg; the question gives weight in newtons.

4. B — Speed 1 mark

Kinetic energy $E_k = \frac{1}{2}mv^2$. It depends on **mass** and the **square of speed (velocity)**. Height is linked to potential energy, weight is force due to gravity, and pressure is not involved in kinetic energy formula.

5. B — 50 J 1 mark

By conservation of energy (no friction), KE at the ground = PE at maximum height = $\frac{1}{2}mv^2 = \frac{1}{2} \times 4 \times 5^2 = \frac{1}{2} \times 4 \times 25 = \mathbf{50 \text{ J}}$. The energy returns to kinetic form when the ball returns to ground level. This is the work-kinetic energy theorem in action.

6. B — Change forms 1 mark

The **Law of Conservation of Energy** states: "Energy can neither be created nor destroyed in any process; it can be converted from one form to another but the total amount of energy remains constant." Options A and D describe dissipation (a real phenomenon), but the law says energy *changes forms*, not that it is truly lost.

7. C — Watt 1 mark

Power $P = W/t$. Its SI unit is the **watt (W)**, named in honour of **James Watt**, a Scottish physicist who invented the first steam engine. $1 \text{ W} = 1 \text{ J s}^{-1}$. Horsepower is a non-SI unit ($1 \text{ hp} = 746 \text{ W}$).

8. B — 20 W 1 mark

$P = W/t = 100 \text{ J} / 5 \text{ s} = \mathbf{20 \text{ W}}$. 10 W would result from 100 J in 10 s ; 105 W is a common addition error; $500 \text{ W} = 100 \times 5$ (multiplication instead of division).

9. D — 746 W 1 mark

One **horsepower (hp)** = $550 \text{ ft}\cdot\text{lb/s} = \mathbf{746 \text{ W}}$ (exactly). This is a standard conversion from the textbook. 756, 716, and 736 are common distractors from approximate or incorrect conversions.

10. C — Sunlight 1 mark

Sunlight (solar radiation) is a renewable energy resource; it is replenished continuously by the Sun. Coal and natural gas are fossil fuels (non-renewable). Uranium is used as nuclear fuel and is also non-renewable because the material itself (U-235) is finite and not naturally replenished on a human timescale.

11. C — 30% 1 mark

$\eta = (E_{\text{out}}/E_{\text{in}}) \times 100\% = (120/400) \times 100\% = 0.3 \times 100\% = \mathbf{30\%}$. This is Example 6.6 from the textbook. The engine wastes 280 J as heat and other forms of energy, keeping only 120 J as useful work.

12. D – 1 (100%) **1 mark**

A practical engine **cannot** have efficiency equal to or greater than 1 (100%) because frictional forces always dissipate some energy as heat. Work output is always less than work input. This would violate the laws of thermodynamics (a perpetual motion machine concept). Efficiency is always < 1 in the real world.

SECTION B – Short Answer Questions (11 × 3 = 33 Marks)**Q.2(i) – Main Question** **3 marks**

Define work with its SI unit. Write the formula $W = Fd$ and state two conditions when work done is zero.

Definition: Work is said to be done when a force displaces a body in its own direction. **1 mark**

Formula: $W = (F)(d)$, where F is applied force in newtons and d is displacement in metres. SI unit of work is the **joule (J)**; $1 \text{ J} = 1 \text{ N} \cdot \text{m}$. **1 mark**

Two conditions for zero work: **1 mark**

- When displacement is zero (e.g., pushing a wall that does not move).
- When force is perpendicular to displacement (e.g., centripetal force on a satellite in circular orbit does no work since force and displacement are at 90°).

OR

Muhammad Taha pushes a box with 150 N through 5 m. Calculate work done.

Given: $F = 150 \text{ N}$, $d = 5 \text{ m}$ **Required:** $W = ?$

Solution: $W = F \times d = 150 \text{ N} \times 5 \text{ m}$ **1 mark**

$W = 750 \text{ J}$ **2 marks**

Muhammad Taha does 750 J of work on the box.

Q.2(ii) – Main Question **3 marks**

Define kinetic energy. Write and derive the formula $E_k = \frac{1}{2}mv^2$. Is it scalar or vector?

Definition: The energy possessed by a body due to its motion is called **kinetic energy**. **1 mark**

Derivation: A constant force F acts on mass m , starting from rest ($v_i = 0$) to final velocity v . Average velocity $= v/2$; acceleration $a = v/t$. Change in KE = work done $= F \cdot d = (ma)(v_{\text{ave}} \cdot t) = m(v/t)(v/2)(t)$ **1 mark**

$$E_k = \frac{1}{2}mv^2$$

where m = mass (kg), v = speed (m/s). Kinetic energy is a **scalar quantity**. **1 mark**

OR

Babar Azam hits a ball giving it KE = 50 J; mass = 120 g. Find its speed.

Given: $E_k = 50 \text{ J}$, $m = 120 \text{ g} = 0.12 \text{ kg}$ **Required:** $v = ?$

$E_k = \frac{1}{2}mv^2 \Rightarrow v^2 = 2E_k/m = (2 \times 50)/0.12$ **1 mark**

$v^2 = 833.3 \Rightarrow v = \sqrt{833.3}$ **1 mark**

$v = 28.9 \text{ m/s}$ **1 mark**

Q.2(iii) — Main Question 3 marks

Define gravitational potential energy. Write $E_p = mgh$ and explain each symbol. Give two real-life examples.

Definition: The energy possessed by a body by virtue of its position (in a gravitational force field) is called **gravitational potential energy**. 1 mark

Formula: $E_{p,grav} = mgh$ where m = mass (kg), $g = 9.8 \text{ m/s}^2$ (acceleration due to gravity), h = height above reference point (m). 1 mark

Two examples: 1 mark

- A book placed on a shelf has gravitational PE relative to the floor.
- Water stored at height in a dam reservoir has gravitational PE.

OR

Kamil jumps 1.5 m high; mass 60 kg. Calculate his gravitational PE at the top.

Given: $m = 60 \text{ kg}$, $h = 1.5 \text{ m}$, $g = 9.8 \text{ m/s}^2$ **Required:** $E_p = ?$

$E_p = mgh = 60 \times 9.8 \times 1.5$ 1 mark

$E_p = 60 \times 14.7$ 1 mark

$E_p = 882 \text{ J}$ 1 mark

This gravitational PE came from the kinetic energy of his upward push.

Q.2(iv) — Main Question 3 marks

State the Law of Conservation of Energy. Using a falling ball, explain the KE/PE interchange.

Law: "Energy can neither be created nor destroyed in any process. It can be converted from one form to another, but the total amount of energy remains constant." Mathematically: $E_{Ti} = E_{Tf}$. 1 mark

Falling ball example: 2 marks

- **At top (initial point):** $E_k = 0$, $E_p = mgh$. Total energy = mgh .
- **During fall:** As height decreases, E_p decreases; E_k increases. At any point: $E_k + E_p = mgh$ (constant).
- **Just before ground (final point):** $E_p = 0$, $E_k = mgh$. Total energy = mgh . Energy is conserved throughout.

OR

Football $m = 0.45 \text{ kg}$ kicked at 22 m/s . Find maximum height (no air resistance). $g = 9.8 \text{ m/s}^2$.

Given: $m = 0.45 \text{ kg}$, $v = 22 \text{ m/s}$, $g = 9.8 \text{ m/s}^2$

Using conservation: $\frac{1}{2}mv^2 = mgh \Rightarrow h = v^2/(2g)$ 1 mark

$h = (22)^2 / (2 \times 9.8) = 484/19.6$ 1 mark

$h = 24.7 \text{ m}$ 1 mark

Q.2(v) — Main Question 3 marks

Explain energy conversion with two examples: (a) dam and (b) a cyclist going up a hill.

Energy conversion refers to the process of changing energy from one form to another. 1 mark

(a) Dam (Hydroelectric generation): Water stored at height has **gravitational PE**. When it falls on turbines, PE converts to **KE** of water, then to **electrical energy** via generators. 1 mark

(b) Cyclist going uphill: The cyclist uses stored **chemical PE** in body to do work against gravity. At the top, energy is stored as **gravitational PE**. Going downhill, PE converts back to **KE**, allowing the cyclist to ride without pedalling. 1 mark

OR

Why is energy always dissipated during conversion? Where does the 80% go in a petrol car engine?

During every energy conversion, some energy is converted into forms that do not serve the intended purpose, typically **heat**. Scientists call this **dissipated energy**. 1 mark

In an automobile driven with petrol, only **17 to 21%** of chemical energy moves the car. 1 mark

The remaining ~80% is lost to: **heat generated by the engine, friction in moving parts**, and exhaust gases. Electric cars are far more energy-efficient than internal combustion engine cars because they convert electrical energy to motion with far less waste. 1 mark

Q.2(vi) — Main Question 3 marks

Differentiate between renewable and non-renewable energy resources. Give two examples of each.

Renewable	Non-Renewable
Replaced rapidly by natural processes; cannot simply run out (technology not fuel). 1 mark	Limited; cannot be replaced naturally in a short time. 1 mark
Examples: Sunlight (solar), Wind energy	Examples: Coal, Natural gas

Renewable sources are better for the environment as they do not emit greenhouse gases during operation. 1 mark

OR

What is a perpetual motion machine? Why is it impossible? Name the two laws it violates.

Perpetual motion is the motion of bodies that continues forever in an unperturbed system. A **perpetual motion machine** is a hypothetical machine that can do work indefinitely without an external energy source. 1 mark

It is **impossible** because every real process dissipates some energy as heat due to friction and other losses. Work output is always less than work input. 1 mark

It would violate the **First Law of Thermodynamics** (energy cannot be created) and/or the **Second Law of Thermodynamics** (entropy always increases; heat cannot be fully converted to work). 1 mark

Q.2(vii) — Main Question 3 marks

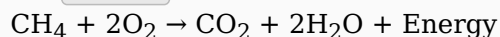
How are fossil fuels formed? Name the three main types. Write the combustion equation for CH₄.

Formation: Fossil fuels are the remains of millions-of-years-old plant life (now coal) or aquatic animal life (now gasoline and natural gas). Chemical energy stored in fossil fuels is obtained through combustion.

1 mark

Three types: **Coal** (most abundant fossil fuel globally), **Oil/Petroleum** (crude oil refined into gasoline, jet fuel, heating oil), **Natural Gas** (primarily methane, a byproduct of oil extraction). 1 mark

Combustion equation for methane: 1 mark



OR

Explain nuclear fuel. Name two materials. State one advantage and one disadvantage.

Nuclear fuel: Nuclear power plants harness energy in the nucleus of an atom through **nuclear fission** (splitting heavy atoms). This produces large amounts of thermal energy used to boil water, generate steam, and drive turbines to produce electricity. 1 mark

Two materials: **Uranium-235 (U-235)** (primary nuclear fuel) and **Plutonium-239** (produced by irradiating U-238 in reactors). 1 mark

Advantage: Small amount of fuel produces a large amount of energy; does not pollute the air or emit greenhouse gases during operation. **Disadvantage:** Produces **radioactive waste** which is highly toxic and causes burns, cancers, blood diseases, and bone decay. 1 mark

Q.2(viii) — Main Question 3 marks

Explain hydroelectric generation. What energy conversion takes place? Why can large reservoirs harm ecosystems?

Hydroelectric generation uses the **gravitational potential energy of water** stored at height (in a reservoir behind a dam) to generate electricity. 1 mark

Energy conversion: Gravitational PE of water → KE (as water falls) → Rotational KE of turbines → Electrical energy (via generator). 1 mark

Ecosystem harm: Large reservoirs flood vast areas of farmland, significantly changing the ecosystem. The full consequences on animals, plants and habitats may not be predictable. Smaller generation facilities are now becoming more popular due to technological advances. 1 mark

OR

Describe solar radiation as an energy source. Two methods of harnessing. One advantage, one disadvantage.

Solar radiation: Energy from direct sunlight can be used to produce electricity. Today solar cells power everything from calculators to small cities. The Sun is a completely free and renewable energy resource.

1 mark

Two methods: (1) **Solar photovoltaic cells (solar cells):** semiconductor materials (silicon) convert sunlight directly into electricity. (2) **Solar thermal systems:** Concentrated Solar Power (CSP) systems use mirrors/lenses to concentrate sunlight, heat a fluid, produce steam, and drive turbines. 1 mark

Advantage: Clean, renewable, emits no greenhouse gases during operation. **Disadvantage:** Intermittent and variable (depends on weather and time of day); significant land area required. 1 mark

Q.2(ix) — Main Question 3 marks

Define power. Write $P = W/t$ and state its SI unit. Convert 2 hp to watts.

Definition: Power is the time rate at which work is done or the rate at which energy is converted from one form to another. 1 mark

Formula: $P = W/t = E/t$, where W is work in joules and t is time in seconds. 1 mark

SI unit: Watt (W), where $1 \text{ W} = 1 \text{ J s}^{-1}$, named after James Watt.

Conversion: $2 \text{ hp} = 2 \times 746 \text{ W} = 1492 \text{ W}$ 1 mark

OR

Tesla car engine: $W = 3.69 \times 10^7 \text{ J}$ in 1 minute. Find power in watts and in hp.

Given: $W = 3.69 \times 10^7 \text{ J}$, $t = 1 \text{ min} = 60 \text{ s}$ **Required:** P in W and hp

$P = W/t = (3.69 \times 10^7) / 60$ 1 mark

$P = 6.15 \times 10^5 \text{ W} = 615 \text{ kW}$ 1 mark

In hp: $P = 6.15 \times 10^5 / 746 = 825 \text{ hp}$ 1 mark

Q.2(x) — Main Question 3 marks

Define efficiency. Write $\eta = E_{\text{out}}/E_{\text{in}} \times 100\%$. Why can no machine have 100% efficiency?

Definition: Efficiency (η) is the ratio of useful energy output (or work output) to the total energy input (or work input). 1 mark

Formula: $\eta = E_{\text{out}}/E_{\text{in}} \times 100\%$ (efficiency has no unit, as it is a ratio of same quantities) 1 mark

Why not 100%: Frictional forces always dissipate some energy as heat, which is not the desired output. Work output is always less than work input. It is not possible to have a machine with 100% efficiency because some input energy is always wasted as heat, which cannot be avoided. 1 mark

OR

2 hp motor, input energy 2238 J, lifts 10 kg mass to 10 m. Find output energy and efficiency.

Given: $E_{\text{in}} = 2238 \text{ J}$, $m = 10 \text{ kg}$, $h = 10 \text{ m}$, $g = 9.8 \text{ m/s}^2$

Output energy (useful work = lifting the load):

$E_{\text{out}} = mgh = 10 \times 9.8 \times 10 = 980 \text{ J}$ 1 mark

Efficiency: $\eta = E_{\text{out}}/E_{\text{in}} \times 100\% = (980/2238) \times 100\%$ 1 mark

$\eta = 43.8\%$ 1 mark

Q.2(xi) — Main Question 3 marks

A car has $KE = E_k$. By what factor does KE change if velocity is doubled? Show mathematical reasoning.

Original KE: $E_k = \frac{1}{2}mv^2$ 1 mark

New velocity: $v' = 2v$. New KE: $E_k' = \frac{1}{2}m(2v)^2 = \frac{1}{2}m \times 4v^2 = 4(\frac{1}{2}mv^2)$ 1 mark

$E_k' = 4E_k$

The kinetic energy becomes **4 times** the original value when velocity is doubled, because KE depends on the *square* of velocity. 1 mark

OR

5 kg steel ball dropped from 12 m tower. Find KE on hitting ground and speed. ($g = 9.8 \text{ m/s}^2$)

Given: $m = 5 \text{ kg}$, $h = 12 \text{ m}$, $g = 9.8 \text{ m/s}^2$

By conservation of energy: $E_k = E_p = mgh$ 1 mark

$E_k = 5 \times 9.8 \times 12 = 588 \text{ J}$ 1 mark

Speed: $\frac{1}{2}mv^2 = 588 \Rightarrow v^2 = 2 \times 588/5 = 235.2 \Rightarrow v = \sqrt{235.2} = 15.34 \text{ m/s}$ 1 mark

Q.3(i) — Main Question 5 marks

Define work and its SI unit. State two zero-work conditions. Derive $E_k = \frac{1}{2}mv^2$ graphically. State the work-energy theorem. A 2.0 kg rock falls 15 m of a 20 m building: find KE and remaining PE. ($g = 9.8 \text{ m/s}^2$)

1. Work (Definition and unit): Work is done when a force displaces a body in its own direction: $W = F \times d$. SI unit: **joule (J)**; $1 \text{ J} = 1 \text{ N}\cdot\text{m}$. 1 mark

2. Two conditions for zero work: (i) Displacement = 0 (body does not move). (ii) Force perpendicular to displacement (e.g., centripetal force on a satellite). 1 mark

3. Derivation of $E_k = \frac{1}{2}mv^2$ (graphical method): 1 mark

Consider a constant force F on mass m , starting from rest ($v_i = 0$), reaching velocity v over displacement d . From the force-displacement graph, the area under the graph is a rectangle:

$$\text{Work done} = \text{Area} = F \times d$$

Since average velocity $v_{\text{avg}} = v/2$ and acceleration $a = v/t$, displacement $d = v_{\text{avg}} \times t = (v/2)(t)$. Also $F = ma = m(v/t)$. Therefore:

$$W = F \times d = m(v/t) \times (v/2)(t) = \frac{1}{2}mv^2$$

$$E_k = \frac{1}{2}mv^2$$

4. Work-Energy Theorem: The work done on an object equals the change in its kinetic energy: $W = \Delta E_k$.

1 mark

5. Numerical (2.0 kg rock, $h = 20 \text{ m}$ total, fallen 15 m): 1 mark

Height fallen so far = 15 m. Remaining height = $20 - 15 = 5 \text{ m}$.

KE gained = $mgh_{\text{fallen}} = 2.0 \times 9.8 \times 15 = \mathbf{294 \text{ J}}$

Remaining PE = $mgh_{\text{remaining}} = 2.0 \times 9.8 \times 5 = \mathbf{98 \text{ J}}$

Check: Total = $294 + 98 = 392 \text{ J} = mgh_{\text{total}} = 2.0 \times 9.8 \times 20 = 392 \text{ J}$. ✓ Conserved.

OR

Define gravitational PE and derive $E_p = mgh$. 5 kg ball dropped from 12 m: (a) PE at top, (b) KE and speed at ground, (c) work done against air resistance if only 540 J KE at ground.

Gravitational PE (definition): Energy possessed by a body due to its position in a gravitational field. A book on a shelf, water in a dam reservoir are classic examples. 1 mark

Derivation of $E_p = mgh$: Consider mass m lifted vertically by force F through height h . To lift at constant speed: $F = W = mg$ (equals weight). Work done against gravity: $W = F \times S = mg \times h = mgh$. This work is stored as gravitational PE. 1 mark

$$E_{p,\text{grav}} = mgh$$

(a) PE at top: $E_p = mgh = 5 \times 9.8 \times 12 = \mathbf{588 \text{ J}}$ 1 mark

(b) KE and speed at ground (conservation, no friction): 1 mark

$$E_k = E_p = 588 \text{ J} \quad \frac{1}{2}mv^2 = 588 \Rightarrow v = \sqrt{(2 \times 588/5)} = \sqrt{235.2} = \mathbf{15.34 \text{ m/s}}$$

(c) Work done against air resistance: 1 mark

$$W_{\text{friction}} = E_p - KE_{\text{ground}} = 588 - 540 = \mathbf{48 \text{ J}}$$

Air resistance did 48 J of work converting mechanical energy to heat.

Q.3(ii) — Main Question 5 marks

State the Law of Conservation of Energy with its mathematical form. Show $E_k + E_p = \text{constant}$ for a falling ball. Football $m = 0.45 \text{ kg}$, kicked at 22 m/s : (a) max height, (b) work against air resistance if only reaches 20 m .

Law of Conservation of Energy: "Energy can neither be created nor destroyed in any process. It can be converted from one form to another, but the total amount of energy remains constant." 1 mark

Mathematically: $E_{Tf} = E_{Ti}$ (Total final energy = Total initial energy). If $W = 0$ then $\Delta E = 0$.

Falling ball proof ($E_k + E_p = \text{constant}$): 2 marks

INITIAL POINT (height h above ground):

$$E_k = 0 \quad (\text{at rest})$$

$$E_p = mgh$$

$$E_{\text{Total}} = mgh$$

MIDWAY (height $h/2$):

$$E_p = mg(h/2)$$

$$E_k = mg(h/2) \quad (\text{gained by losing PE})$$

$$E_{\text{Total}} = mgh \quad (\text{unchanged})$$

JUST BEFORE GROUND ($h = 0$):

$$E_p = 0$$

$$E_k = mgh \quad (\text{all PE converted to KE})$$

$$E_{\text{Total}} = mgh \quad (\text{unchanged})$$

At every point: $E_k + E_p = mgh = \text{constant}$. When friction acts: $E_k + W_{\text{friction}} = E_{p,\text{initial}}$.

Numerical — Football ($m = 0.45 \text{ kg}$, $v = 22 \text{ m/s}$, $g = 9.8 \text{ m/s}^2$):

(a) Maximum height (no friction): 1 mark

$$\frac{1}{2}mv^2 = mgh \Rightarrow h = v^2/(2g) = (22)^2/(2 \times 9.8) = 484/19.6 = 24.7 \text{ m}$$

(b) Work done against air resistance (rises only 20 m): 1 mark

$W_{\text{friction}} = E_{p,\text{actual}} - E_{p,\text{without friction}} \dots$ or directly:

$$W_{\text{fr}} = mgh_{\text{actual}} - \frac{1}{2}mv^2 = (0.45 \times 9.8 \times 20) - (0.5 \times 0.45 \times 484)$$

$$W_{\text{fr}} = 88.2 - 108.9 = -20.7 \text{ J} \quad (\text{negative sign indicates friction opposes motion})$$

So 20.7 J of the available 108.9 J was lost to air resistance.

OR

Explain energy conversion and dissipation with three examples. Where does the 80% go in a petrol car? Why are electric cars more efficient?

Energy conversion refers to changing energy from one form to another. During conversion, some energy is lost as **dissipated energy** (heat, sound, etc.) that does not serve the intended purpose. 1 mark

Three examples: 2 marks

System	Energy Conversion
Dam (Hydroelectric)	Gravitational PE of water $\rightarrow$ KE of water $\rightarrow$ electrical energy via turbine & generator
Cyclist going uphill	Chemical PE in body $\rightarrow$ KE of pedalling $\rightarrow$ gravitational PE at top of hill
Petrol car engine	Chemical PE in fuel $\rightarrow$ KE of car + heat (waste) + friction losses

Where does the 80% go in a petrol car? Only 17–21% of chemical energy moves the car. The remaining ~80% is lost as: **heat from the engine, exhaust gases, and friction in mechanical parts**. Scientists call this "dissipated energy." 1 mark

Electric cars are far more energy-efficient because they directly convert electrical energy to mechanical motion using electric motors, which have much lower friction and heat losses compared to internal combustion engines. 1 mark

Q.3(iii) — Main Question 5 marks

Define power and write $P = W/t$. Explain $1 \text{ W} = 1 \text{ J s}^{-1}$. Define kWh and show $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$. Find mass M lifted by 1960 W lifter ($h = 2 \text{ m}$, $t = 3 \text{ s}$). Convert 1960 W to hp.

1. Power (definition): Power is the time rate at which work is done or energy is converted. $P = W/t = E/t$.

1 mark

SI unit: Watt (W), named after **James Watt**, Scottish physicist who invented the first steam engine.

$1 \text{ W} = 1 \text{ J s}^{-1}$ means 1 watt is the power when 1 joule of work is done per second.

2. Kilowatt-hour (kWh): A unit of energy used in electricity billing. It is the energy converted or consumed at the constant rate of 1 kW for 1 hour. 1 mark

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3,600,000 \text{ J} = 3.6 \times 10^6 \text{ J}$$

The electricity bills we pay are measured in kWh (1 unit = 1 kWh).

3. Finding mass M (weight lifter): 2 marks

Given: $P = 1960 \text{ W}$, $h = 2 \text{ m}$, $t = 3 \text{ s}$, $g = 9.8 \text{ m/s}^2$

Work done = $P \times t = 1960 \times 3 = 5880 \text{ J}$ 1 mark

Also $W = mgh \Rightarrow M = W/(gh) = 5880/(9.8 \times 2) = 5880/19.6$

$M = 300 \text{ kg}$ 1 mark

4. Convert 1960 W to hp: 1 mark

$P = 1960/746 = 2.63 \text{ hp}$

OR

Define efficiency and write formula. Using Table 6.3: why is incandescent bulb 5% and fluorescent 20% efficient? Why does 100% efficiency violate thermodynamics? Calculate efficiency of 2 hp motor lifting 10 kg to 10 m in 1.5 s.

Efficiency (η): Ratio of useful energy (work) output to total energy (work) input. 1 mark

$$\eta = E_{\text{out}}/E_{\text{in}} \text{ (as fraction)} \quad \text{or} \quad \eta\% = (E_{\text{out}}/E_{\text{in}}) \times 100\%$$

Also: $P_{\text{out}} = \eta \times P_{\text{in}}$

Incandescent vs fluorescent: 1 mark

An incandescent bulb is designed to produce light but only 5% of electrical energy becomes light; the remaining 95% becomes waste thermal energy (heat). A fluorescent lamp converts 20% to light and wastes only 80% as heat. Fluorescent lamps are therefore 4 times more efficient.

Why 100% efficiency is impossible: Frictional forces always dissipate some energy as heat. Work output is always less than work input. A 100% efficient system would violate the **Second Law of**

Thermodynamics (a perpetual motion machine). 1 mark

Numerical (2 hp motor, 1.5 s, lifts 10 kg to 10 m):

(a) Input energy: $2 \text{ hp} = 2 \times 746 = 1492 \text{ W}$; $E_{\text{in}} = P \times t = 1492 \times 1.5 = 2238 \text{ J}$ 1 mark

(b) Output energy: $E_{\text{out}} = mgh = 10 \times 9.8 \times 10 = 980 \text{ J}$

(c) Efficiency: $\eta = (980/2238) \times 100\% = 43.8\%$ 1 mark

Q.3(iv) — Main Question 5 marks

Differentiate renewable/non-renewable energy with a table (3 examples each). Describe fossil fuels (formation, extraction, combustion equation for CH_4). Two environmental problems. Why must society shift to renewable sources?

1. Renewable vs Non-Renewable (Table): 2 marks

Feature	Renewable	Non-Renewable
Definition	Replaced rapidly by natural processes; cannot simply run out	Limited; cannot be naturally replaced in a short time
Examples	Solar radiation, Wind energy, Hydroelectric (water)	Fossil fuels (coal, oil, natural gas), Nuclear fuel (uranium)
Environmental impact	Generally cleaner; minimal greenhouse gases during operation	Combustion releases CO_2 and other pollutants

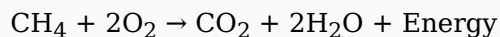
2. Fossil Fuels: 2 marks

Formation: Fossil fuels are the remains of millions-of-years-old plant life (now coal) or aquatic animal life (now gasoline/natural gas). They are hydrocarbons: molecules made up of hydrogen and carbon atoms.

Three types: (A) **Coal:** most abundant fossil fuel globally; estimated reserve of one million metric tonnes; causes significant atmospheric pollution. (B) **Oil (Petroleum):** crude oil refined into gasoline, jet fuel, and heating oil; favoured for high energy output per unit. (C) **Natural Gas:** primarily methane; often a byproduct of oil extraction; easily transported.

Extraction: Coal is mined; oil and natural gas are drilled from reservoirs. Extracted fuels are refined/processed to remove impurities before use.

Combustion equation for methane:



3. Two environmental problems: 1 mark

- Combustion releases **carbon dioxide (CO₂)** and other harmful substances into the atmosphere, causing **air pollution and climate change (global warming)**.
- Burning fossil fuels also produces **greenhouse gases** that trap heat, leading to rising global temperatures.

Why shift to renewables: Society cannot persistently consume increasing amounts of non-renewable energy without taking future generations into account. Renewable sources do not diminish over time, do not harm the biosphere, and are better for the environment. The challenge lies in developing sustainable energy sources and processes.

OR

Describe three renewable energy resources (hydroelectric, solar, wind): energy conversion, electricity generation, one advantage and one disadvantage each. Then: find time for 8.13 kW pump to lift 2500 kg of water 100 m. ($g = 9.8 \text{ m/s}^2$)

1. Hydroelectric Energy: 1 mark

Conversion: Gravitational PE of water $\rightarrow$ KE $\rightarrow$ electrical energy.

Generation: Water stored at height in a reservoir behind a dam falls through intakes, spins turbines connected to generators, producing electricity.

Advantage: Renewable and highly efficient (95%); once built, minimal environmental operation impact.

Disadvantage: Large reservoirs flood farmland and change ecosystems.

2. Solar Radiation: 1 mark

Conversion: Solar (radiant) energy $\rightarrow$ electrical energy (via photovoltaic cells) or thermal energy (via solar thermal systems).

Generation: Solar photovoltaic cells (silicon semiconductors) convert sunlight directly into electricity. CSP systems concentrate sunlight to heat a fluid, produce steam, drive turbines.

Advantage: Completely free, renewable, emits no greenhouse gases. **Disadvantage:** Intermittent (depends on sunlight); significant land area required.

3. Wind Energy: 1 mark

Conversion: KE of wind $\rightarrow$ rotational KE of turbine blades $\rightarrow$ electrical energy.

Generation: Wind turbines have large rotor blades connected to a hub, gearbox, and generator. Blades spin due to wind, the generator produces alternating current (AC). Wind farms group turbines in areas with consistent strong winds.

Advantage: Eco-friendly, renewable, zero greenhouse gas emissions during operation. **Disadvantage:** Only viable in areas with constant strong winds; infrastructure is expensive.

4. Numerical — Pump: 2 marks

Given: $P = 8130 \text{ W}$ (8.13 kW), $m = 2500 \text{ kg}$, $h = 100 \text{ m}$, $g = 9.8 \text{ m/s}^2$

Work needed: $W = mgh = 2500 \times 9.8 \times 100 = 2,450,000 \text{ J}$ 1 mark

Time: $t = W/P = 2,450,000/8130 = 301.4 \text{ s} \approx 5 \text{ minutes}$ 1 mark

(Cross-check: The textbook numerical answer for pumping 2500 kg to 100 m in 5 min = 300 s requires power = $2,450,000/300 = 8167 \text{ W} \approx 8.13 \text{ kW}$. Result confirmed.)